Field Hours

CSCE 436 Computer-Human Interaction

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 50 minutes

A task, not a screen

Start from a task a specific person does, in their words. 'Look up today's patient before they reach the desk' is a task. 'A modern dashboard' is a mood. The person, the place, and the success condition come before any drawing.

Write five tasks. If you cannot, you are designing for yourself. CSCE 655 will run a study. This course is the first time you watch the task instead of the widget.

Two descriptions of one morning.

A screen

Cards, a chart, a search box

No person, no deadline

A task

The clerk, 30 seconds, the right chart

Success is the chart in hand

The front desk

1. Person: the clerk, standing, one hand free.

2. Task: identify the patient who just walked in, from a spoken last name, before the line grows.

3. Success: the correct row is visible without a second question. Failure: the clerk asks the patient to spell it while the line waits.

Problem 1. You shipped a dashboard the clerk never opens. Did you finish the task?

Answer. No. The task was identifying the patient at the desk. A screen nobody uses during that task did not change the morning. Go watch the task before you add another chart.

Problem 2. Why is this course, or CSCE 444, named before CSCE 655?

Answer. 655 assumes you have already named a user and a task and watched someone struggle. The catalog accepts 436 or 444 for that preparation. This site teaches it as 436. Take one.

Lecture 2 · 50 minutes

A prototype you can throw away

The first artifact should be cheap to discard. Paper, or a few static screens, is enough to walk a task. A working

database at this stage makes you defend the wrong decisions, because they were expensive.

Walk the five tasks on the paper. Where the person hesitates, the prototype is wrong or the task is wrong. Change the paper. Do not explain the hesitation away.

One pass.

1. Five tasks

2. A paper screen for each step

3. One person walks them

4. Change the paper, not the speech

The spoken name

1. The paper shows a single field and a list.

2. The clerk says the name and the walker types it wrong on purpose. The list is empty.

3. The next paper shows close matches without a second click. That change took a pencil. The same change in a finished app would have been a negotiation.

Problem 1. The prototype looks unfinished and the stakeholder is embarrassed. What do you protect?

Answer. The right to throw it away. A polished look at this stage hides the hesitations. Invite the criticism while the artifact is still paper.

Problem 2. When is a paper prototype the wrong tool?

Answer. When the question is about timing, animation, or a hardware constraint the paper cannot fake. Then build the smallest running slice of that one question, not the whole product.

Lecture 3 · 50 minutes

A session

A session is one person, who did not build the thing, doing the tasks while you stay quiet. You record what they did, not what they said they would do. Five people will show you the broken task. They will not give you a percentage you can publish. Say the sample size.

Ask them to think aloud if they can. Do not help on the first failure. Helping teaches you nothing about the next clerk who will not have you standing there. Note the time and the exact miss.

A small session.

Read the task aloud -> Stay quiet -> Write the miss -> Thank them -> Change one thing

Three clerks

1. Task: find Nguyen from a spoken name, thirty seconds.

2. Two of the three type the name and freeze on an empty list. One tries a first name and also fails.

3. You do not average this into a satisfaction score. You write: empty exact-match results stranded two

people. The next build must show near matches.

Problem 1. You explained the icon and then the person succeeded. What can you claim?

Answer. That they can do it after your explanation. You cannot claim the icon works. The explanation will not ship. Run the task again without helping, on the changed design.

Problem 2. Is three people a study you can generalize to every clinic?

Answer. No. It is enough to see a repeated miss. CSCE 655 will talk about what a small sample can and cannot support. Do not invent a percentage from three.

Lecture 4 · 50 minutes

Access

A person who does not use a pointer still has the task. Keyboard order, labels that a screen reader can speak, and contrast you can check are part of the task, not a later coat of paint. If the only path is a drag, you excluded someone.

Try the task with the keyboard alone. Every action the mouse can do needs a key path, or an honest reason the task is spatial. 'We will add access later' means the later release is the first one that includes those people.

One task, more than one body.

Pointer

Keyboard only

Text that can be read aloud

Contrast you measured, not a feeling

The list of patients

1. Tab moves through the search, then the rows, in the order a clerk scans them.

2. Enter opens the row that has focus. There is no mouse-only control.

3. The selected row is not indicated by color alone. A marker or a weight change survives a grayscale printout and a common color blindness.

Problem 1. The designer says color-blind users are an edge case. What is the concrete test you still run?

Answer. Complete the task with color removed, or with a simulation of a common deficiency. If the selected row disappears, the design depends on color. Add a second cue. The size of the population is not the test.

Problem 2. A drag-and-drop queue has no keyboard path. Is the task done?

Answer. Not for someone who does not drag. Add a move action on the focused row, or do not claim the queue is usable. The mouse path can stay.

Lecture 5 · 50 minutes

One change

After the session, change one thing that the misses justify. Do not redesign the product. A second session on the new paper tells you whether that one change helped. Two changes at once and you will not know which one mattered.

Write what you refused to change, and why. A request that fights the task belongs in that list. CSCE 655's capstone weeks assume you have done this loop once.

The loop.

1. Misses from the session

2. One change

3. The same tasks again

4. What you refused

Near matches

1. The change: an empty exact search shows the closest names instead of a blank page.

2. The same three tasks, or three new people if you can get them. You watch whether the freeze is gone.

3. You refuse a request to add a stock photo of a clinic. It does not touch the miss. The refusal is one sentence.

Problem 1. You changed the colors, the layout, and the search in one night. The second session went better. What do you know?

Answer. That the bundle went better. You do not know which change did it. Next time change one. If you must ship the bundle, say so, and do not claim you measured the search.

Problem 2. A stakeholder wants the photo. The session did not mention it. Where does it go?

Answer. On the list of things you refused, with the reason: no miss in the session supports it. If they insist, it is a new requirement, not a usability finding.

Lecture 6 · 50 minutes

What the graduate studio assumes

You can name a person and a task, walk a cheap prototype, sit through a quiet session, reach the task without a pointer, and change one thing because of a miss. CSCE 655 will add study design, consent, and a longer argument about what a small sample means.

The catalog also accepts CSCE 444 in this slot. You do not need both. You do need one of them before you treat 655 as a first course in watching users.

Bring these.

Five tasks in the user's words

A prototype you were willing to scrap

Notes from a session with no helping

One change and one refusal

A question you should answer cold

1. The person succeeded after you pointed at the button. What claim is illegal?

2. That the button was findable. Your hand was part of the interface.

3. If that distinction is fuzzy, run one more session before the graduate course.

Problem 1. What will you not claim from five people?

Answer. A percentage that generalizes to every user. You can claim the misses you saw, and that a repeated miss is worth a change. The graduate course tightens that language.

Problem 2. Which course on this site stands in for the catalog's CSCE 436 or CSCE 444 line?

Answer. CSCE 436. Take it before CSCE 655 if you have not already done an equivalent studio. It is outside the 36 credits.